

1) What is a Virtual Warehouse

Virtual warehouse = cluster of compute nodes (resources)

Cluster => group of nodes/machines connected together

Compute Resources => CPU + Memory + Temporary storage

Machine/computer/node

A **virtual warehouse** is Snowflake's compute layer. A logical cluster of compute resources (CPU, memory, temp storage) that executes SQL (queries, DML), Snowpark code, COPY/LOAD operations, and other compute tasks.

Data storage (the tables) is separate from compute in Snowflake;

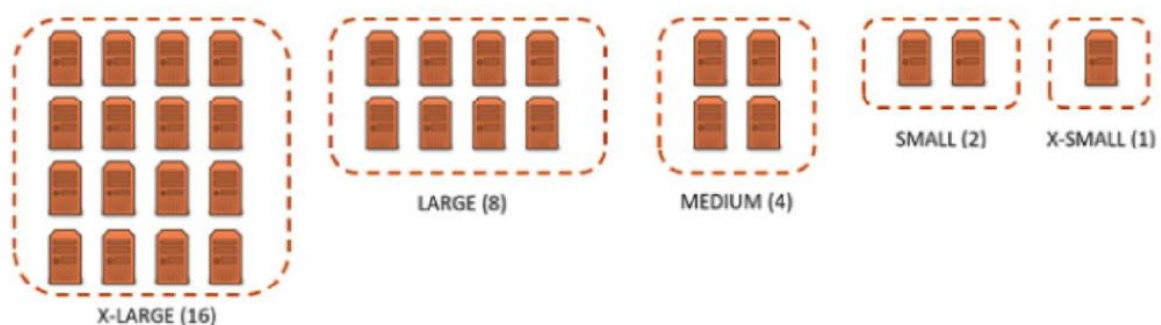
warehouses provide the compute needed to process that data.

Practical consequence: you cannot run queries, load data, or create/refresh materialized views without an active warehouse.

2) Warehouse size — what it is and types

Warehouse size (XSMALL ... 6XLARGE) controls the amount of compute in **each cluster** of a warehouse. Bigger sizes give more cores/memory per cluster and therefore (usually) faster single-query performance. Snowflake supports many sizes (XSMALL, SMALL, MEDIUM, LARGE, XLARGE, 2X–6X-LARGE). The size directly affects credit consumption (cost).

Snowflake Warehouse comes in various T-Shirt sizes



When to choose what:

- Use a **larger size** to speed up heavy/complex queries or very large single loads.
- Use **multiple clusters (scale-out)** for concurrency (many simultaneous queries).

3) What is Auto Suspend and Auto Resume

- **AUTO_SUSPEND**: number of **seconds of inactivity** after which Snowflake automatically **suspends** the warehouse (stops compute and stops incurring credits). Typical value: 60 (1 minute) or 300 (5 minutes).
- **AUTO_RESUME**: when set to TRUE, Snowflake will **automatically resume** the warehouse when a query or other action needs compute. If disabled, someone must manually resume.

Warehouses

New Warehouse
Owner: ACCOUNTADMIN

Type: Standard (Gen1) Size: X-Small

Advanced Options

- ☒ **Auto-resume**
Automatically resumes the warehouse when any statement that requires a warehouse is submitted.
- ☒ **Auto-suspend**
Automatically suspends the warehouse if it is inactive for the specified period of time. [Learn more](#)

Suspend After
1 min(s) of inactivity

1 credit/hour

Cancel Create Warehouse

4) What is a Multi-Cluster Warehouse

A **multi-cluster warehouse** can contain multiple clusters and is used when you need to serve many concurrent users/queries without queuing. Auto-scale mode will add clusters up to the MAX_CLUSTER_COUNT when concurrency increases, and remove them when workload falls. Available in Enterprise+ editions (and above) depending on account.

☒ **Multi-cluster Warehouse**
Scale compute resources as query concurrency needs change. [Learn more](#)

Min clusters	Max clusters	Scaling Policy
1	2	Standard

5) What is Min cluster and Max cluster

When you enable multi-cluster autoscaling you set:

- MIN_CLUSTER_COUNT — minimum clusters to keep running (often 1)
- MAX_CLUSTER_COUNT — maximum clusters Snowflake may start for that warehouse

Snowflake will start/shutdown clusters within that range according to load and the chosen scaling policy.

Here we have 2 types.

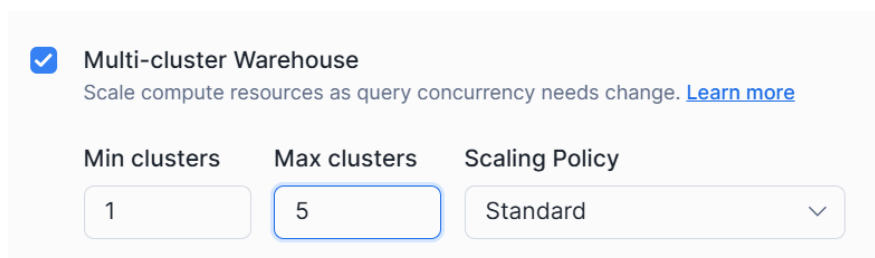
Maximized and Auto-scale mode

1. Auto-Scale mode:

At the time of warehouse creation if we set

Max cluster count (5) > Minimum cluster count (1)

As the query load or concurrent user increases or decreases, the warehouse automatically scale-out/in respectively



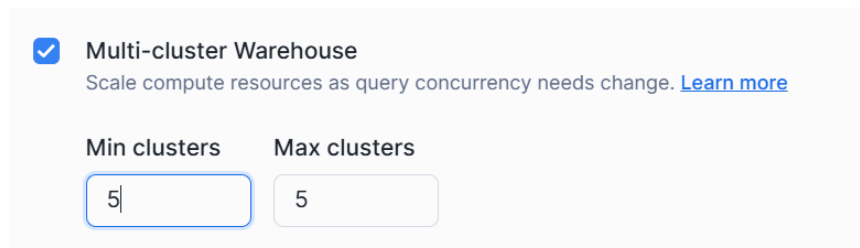
The screenshot shows the 'Multi-cluster Warehouse' configuration interface. It includes a checked checkbox for 'Multi-cluster Warehouse' with a link to 'Learn more'. Below this, there are three input fields: 'Min clusters' with the value '1', 'Max clusters' with the value '5', and 'Scaling Policy' set to 'Standard' with a dropdown arrow.

2. Maximized mode:

Max cluster count =

Minimum cluster count

Snowflake starts all the clusters so that maximum resources are available while the warehouse is running.



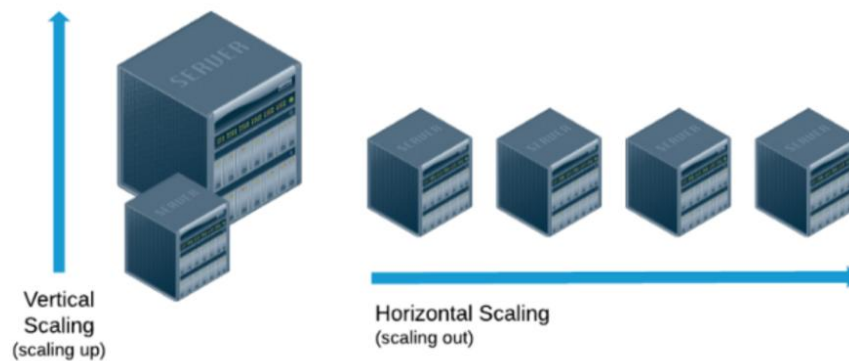
The screenshot shows the 'Multi-cluster Warehouse' configuration interface. It includes a checked checkbox for 'Multi-cluster Warehouse' with a link to 'Learn more'. Below this, there are two input fields: 'Min clusters' with the value '5' and 'Max clusters' with the value '5'.

6) What is Scaling Policy

In Snowflake, we have predefined 2 types of scaling policy.

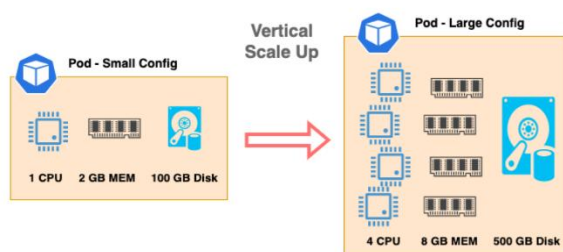
1. **Standard (default policy)** – performance (starts immediately when a query is queued)
2. **Economy** - conserves/saves credits (cluster starts only if it estimates there is enough query load to keep the cluster busy for at least 6 minutes)

7) Vertical and Horizontal scaling in Snowflake



Vertical Scaling (Resizing the warehouse)

Scale up => increasing the warehouse size example from small => M to XL



Purpose is to increase performance of complex queries

An increase in T-Shirt size (XS→XL) => increase in CPU, Memory, Temporary Storage.

Scale down => reducing the warehouse size example from large => small

Scale up and down (Re-sizing warehouse) is manual process

Horizontal Scaling (Adding clusters of same size)

Scale out => adding clusters to a warehouse (Multi cluster warehouse).

Scale Out



Will help resolve concurrency issues. Multiple users accessing your application at the same time, and you don't have resources, and then auto-scaling will help resolve that issue

Concurrency is executing two or more queries at the same time (in parallel) by single/multiple users.

Scale in => removing clusters in a warehouse (Multi cluster warehouse).